On The Gravitational Nature Of Time

Andrew Mueller[1*]

[1*]Independent Researcher, Milwaukee, 53219, WI, USA.

Corresponding author(s). E-mail(s): AndrewMueller414@gmail.com;

Abstract

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. These transformations satisfy existing experimental validations of special relativity while providing a set of 4 equations yielding our peculiar velocity, a resolution to the SNe/CMB velocity discrepancy, an orbital velocity equation that matches observation with only visible mass, and a direct physical mechanism to the equivalence principle.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of this geometry under the assumption that the $\mathbb{R}^3$ space between two arbitrary points dilates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time where the 4th KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame

preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th axis.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event exists at a single point in space-time; a local measurement as $dt \to 0$. The constant nature of 4-vecs inherent to the definition of an event in SR arises naturally from the coordinate transformations that follow, giving d/v that is equivalent as measured in each respective coordinate system.

Let there be two overlapping coordinate systems, **S** and **U**. Let **S** and **U** represent static positions in time, with **U** representing a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{1}$$

where $\Delta t > 0$.

Let there also be two observers, A and B. Let A and B agree that B should travel at some relativistic velocity v' between two arbitrary points in space contained within **S** and **U**, p_1 and p_2. Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

If A and B were to disagree upon the time at which B reaches p_2, this single, instantaneous event (B reaching p_2) would be required to happen more than once. Therefore, we theorize that it is not possible for A and B to disagree upon the time at which B reaches p_2.

Instead, as the additional factor of γ has been demonstrated experimentally on a number of occasions[7–11], we should conclude that both A and B observe B reaching p_2 in $\frac{d}{v}\gamma$. This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \overbrace{\frac{d \cdot \gamma}{v^{(U)}}}^{B} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{2}$$

This produces a model of relativistic dilation that loosely binds the additional γ factor in the $d = vt\gamma$ equivalence to cosmic inflation, where the additional velocity experienced by B results in a more rapid progression along this concurrent density axis leading to a further separation between any two points in **U**.

2 Theoretical Framework

Let us acknowledge the two primary properties of time:

1. A driving rate of change, as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance, and acts only as a mathematical tool: $t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, now denoted y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart.

As we are removing time as a physically meaningful quantity, let us define

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{3}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change. Let $\widehat{v} = v \cdot 1t$, or the velocity in units of distance.

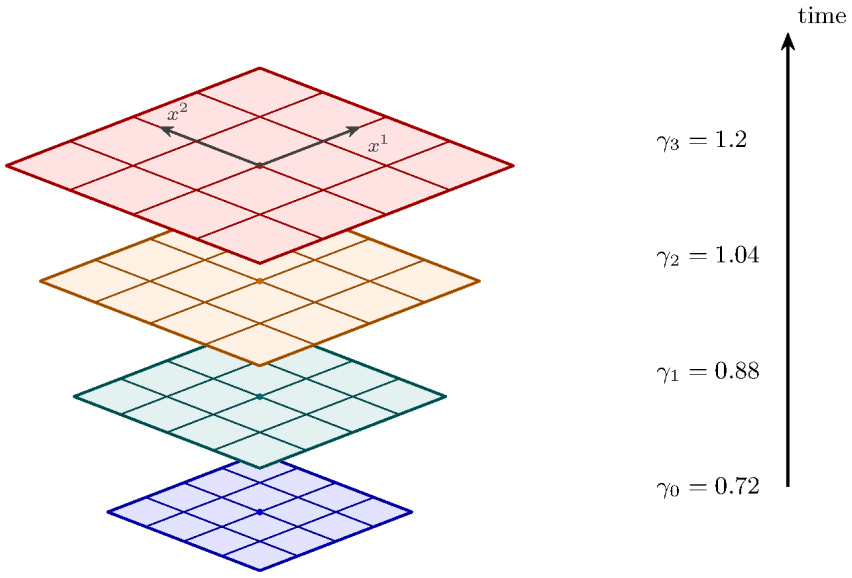


Fig. 1 A plane-based geometric simplification of the concurrent density axis found by these coordinate transformations.

2.1 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.1.1 Earth's Kinematic Velocity From Electromagnetic Parameters

By solving γ for v, let us define $v_{(\gamma)}$ to be

$$v_{(\gamma)} = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{4}$$

And $\delta_{(M,R)}$ to be

$$\delta_{(M,R)} = \frac{d}{dR}g = -2\frac{GM}{R^3} \tag{5}$$

If $\delta_{(M,R)}$ is induced by spatial dilation, we should presume that this dilation is radial (at least in a simplified single body system). We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega_{(M,R)} = \sqrt[3]{\frac{d}{dR}g} = \delta_{(M,R)}^{1/3} \tag{6}$$

If we investigate this quantity for Earth, we find:

$$\omega_{\oplus} = 0.014537\ldots \tag{7}$$

While this approximates 2α, it more closely gives

$$\omega_{\oplus} = \alpha\,(2 - \alpha) \pm 0.0253\% \tag{8}$$

Then by expanding equation 8

$$\alpha\,(2 - \alpha) = \sqrt[3]{\frac{d}{dR}g} \tag{9}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right) = 1.272 \times 10^{21} \tag{10}$$

If we then take the radius independent magnitude, isolating the additional electromagnetic parameters we find

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{11}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 4. This gives

$$\gamma_{\text{em}} = 1 + \frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{12}$$

$$v_{(\gamma_{\text{em}})} = 371,721 \ \frac{m}{s} \tag{13}$$

matching the 2018 Plank data[12] to within $\approx 0.52\%$.

2.1.2 Solar ω

If we expand equation 9 we find

$$\alpha = \Gamma_{(\sigma)} \left(\frac{d}{dR} g \right)^{1/3} \tag{14}$$

Where $\Gamma_{(\sigma)}$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 9 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{(\sigma),\oplus} \approx \frac{1}{2-\sigma} \tag{15}$$

where $\sigma_{\oplus} \approx \alpha$.

The Sun, being the dominant local mass with a surface gravitational acceleration almost 30 times greater than the next body in question, would project this spatial vector (moving space itself) more closely to a purely radial projection. This behavior, the manner in which the Sun dominates the local gravitational environment should allow us to approximate a single body solution near the solar surface.

Let us then inspect $\Gamma_{(\sigma),\odot}$, under the assumption that this quantity acts *inside* of the divergence, acting on each individual $\mathbb{R}^3$ axis.

$$\Gamma_{(\sigma),\odot} \propto \sqrt[3]{\Gamma_{(\theta)}} \tag{16}$$

If we then presume that α in $\Gamma_{(\theta),\oplus}$ is associated with spin, we can approximate this term to be 0, giving

$$\Gamma_{(\sigma),\odot} \approx \frac{1}{\sqrt[3]{2}} \tag{17}$$

Or

$$\alpha = \Gamma_{(\sigma),\odot} \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{18}$$

We find through observation that for the least mass dense body in the system, $\Gamma_{(\sigma),\hbar} \approx 1$.

This gives a set of 3 equations where

$$\begin{aligned} g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3\,(2-\alpha)^3}{2}\right) && \pm 0.008\% \\ g_{\odot} &= R_{\odot}\alpha^3 && \pm 1.426\% \\ g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3 && \pm 1.134\% \end{aligned} \tag{19}$$

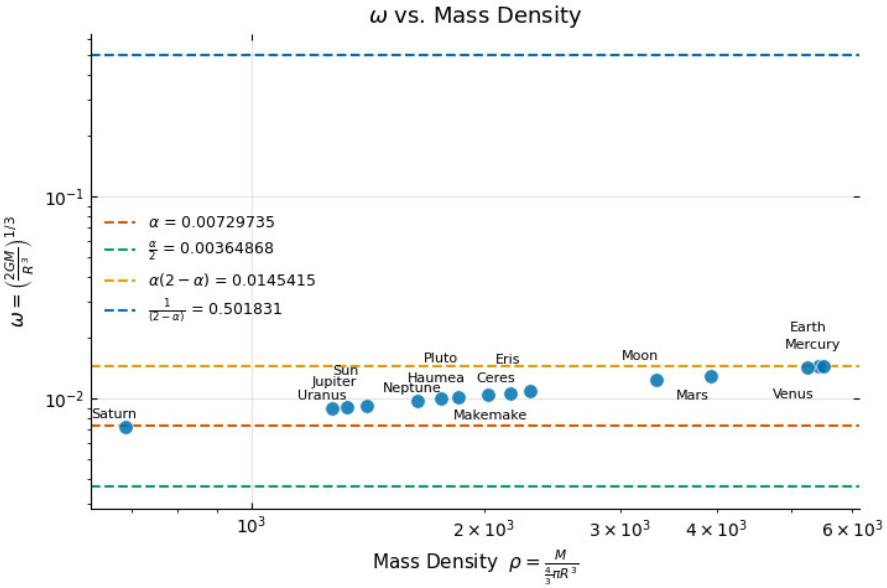


Fig. 2 The inverse of the determinant representing a change in volume, being equivalent to dy, plotted against mass density.

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha\,(2-\alpha)$ as shown in figure 2.

2.2 Gravitational Parameters

2.2.1 Cosmological Velocity

$$\varphi_{(M,R)} = \frac{1}{R}\int \delta_{(M,R)} = 1 + G\frac{M}{R^3} \tag{20}$$

$$v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2_{(M,R)}}} = 525,493.17 \text{ m/s} \tag{21}$$

2.2.2 Kinematic Velocity

$$\int_0^R \int_0^x \left[\delta \frac{1}{v_\gamma} \tau\right] \, dR \, d\tau = 1 + \frac{1}{2} \frac{GM_\oplus}{R_\oplus^3} \left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{22}$$

$$v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2} G \frac{M}{R^3}\right)^2}} = 371,567 \text{ m/s} \tag{23}$$

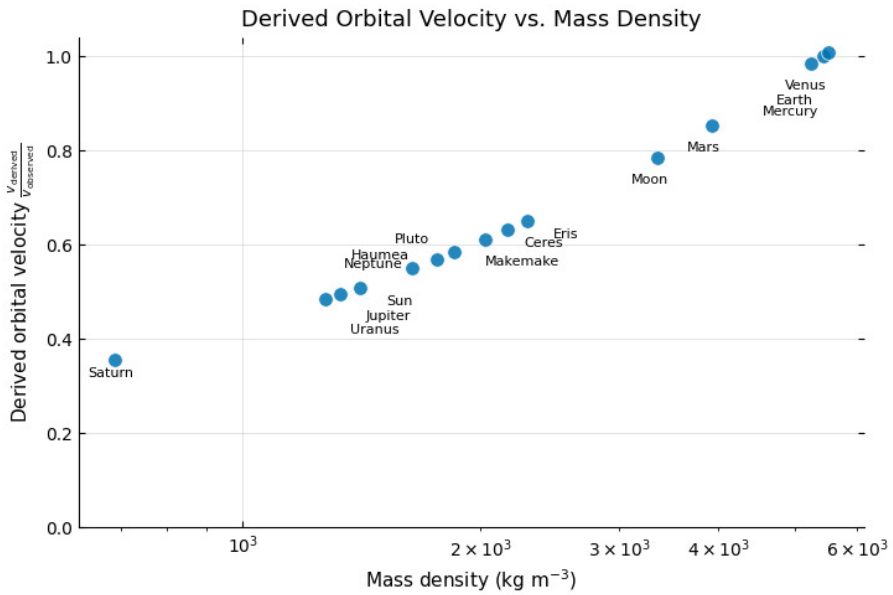


Fig. 3 A trigonometric derivation of a relativistic orbital velocity equation.

2.3 Orbital Velocity

2.3.1 Relativistic Stable Orbit Velocity

$$R_0^2 + \dot{x}^2 dt^2 = \left(\int_0^{R_0} 2G \frac{M}{R^3}\right)^2 = (R_0 + g \, dt)^2 \tag{24}$$

$$v_{\text{orbit}} = \frac{\sqrt{GM\left(GM + 2R^3\right)}}{R^2} = 230,145 \text{ m/s} = v_{\text{eilers}} \pm 0.498\% \tag{25}$$

2.3.2 Kinematic Velocity From Triginometric Orbit Equations

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{26}$$

$$v_{(-\mathrm{dt})} = 371,580 \text{ m/s} \tag{27}$$

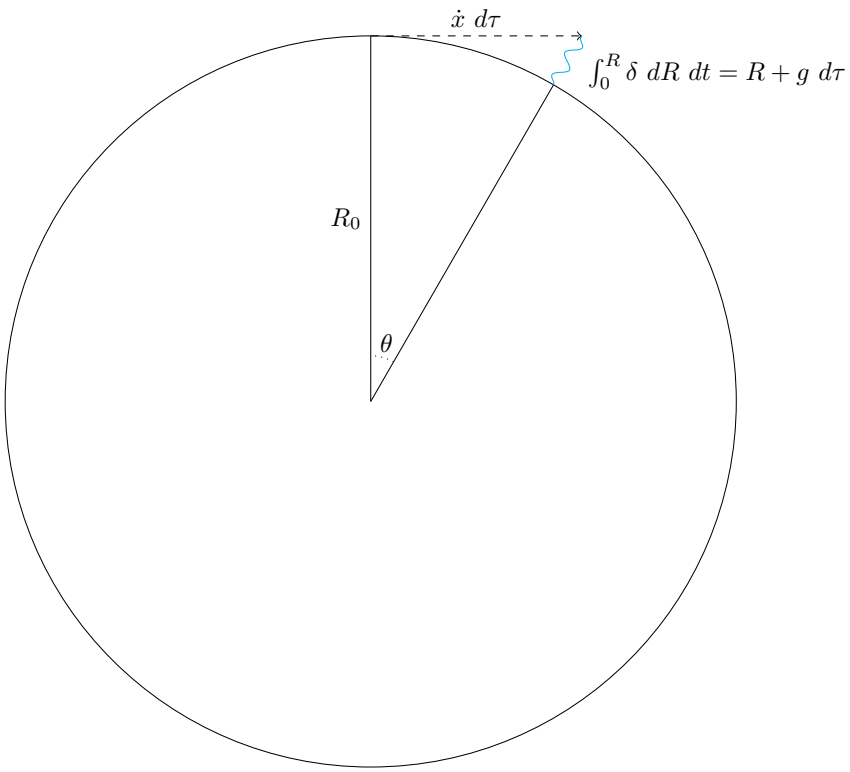


Fig. 4 A trigonometric derivation of a relativistic orbital velocity equation.

3 Results

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5 Equations

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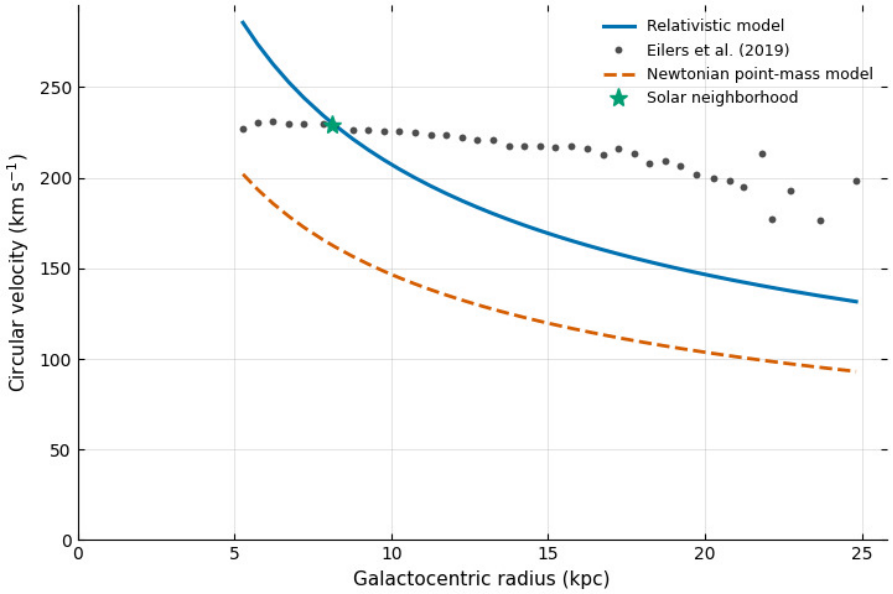


Fig. 5 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

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